Connor Espenshade

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Systems researcher focused on ML infrastructure and inference efficiency, with first-author research and industry experience at Cerebras.

Education

Columbia University, Candidate for BS in Computer Engineering — Sept 2022 – May 2026

- GPA: 3.8/4.0 (Dean's List, Transcript Link)
- Graduate Coursework: Computer Architecture, Operating Systems, Machine Learning, Embedded Systems, System-on-a-Chip (SoC) Platforms, Artificial Intelligence, Natural Language Processing, Computer Vision II
- PhD-level Research Seminars: Embedded Scalable Platforms, Machine Learning Frontiers

Experience

Cerebras Systems, Machine Learning Runtime Engineer Intern – Sunnyvale, CA — May 2025 – Aug 2025

- Delivered C++ wafer-scale simulator to model inter-die connections at region/core/wire granularity with backpressure (capacity, delays) to capture realistic throughput and stall behavior
- Accelerated iteration time 288x from 24h to 5m, enabling rapid validation of new models and algorithms
- Designed diagnostic tests to surface performance bottlenecks and communication stalls across large-scale models
- Modeled region-specific KV-cache behavior to diagnose token drops, batching inefficiencies, and race conditions

Columbia-IBM Sustainable Computing in AI Initiative, Student Lead – New York, NY — Jan 2023 – Present

- Led IBM engineers, Columbia faculty as first author, setting research direction & building most experiments
- Leading research on Escalator, a zero-cost routing system to escalate queries from Llama 3.1 8B to 70B. Training RL reward model for draft embeddings, achieving 87% accuracy to reduce compute costs | Draft Report Link
- Delivered 42% energy savings, 55% latency reduction by collocating PyTorch implementations (GPT-2, BERT, ResNet) on heterogeneous NVIDIA MIG "mini-GPUs" | Publication Link
- Built profiling suite for Llama 3.1 8B, quantified LoRA rank throughput, energy, memory tradeoffs and achieved up to 2.7x efficiency gains with minimal impact to model quality (perplexity) | Publication Link

American Industrial Partners, Data Scientist Intern – New York, NY — July 2024 – Oct 2024

- Integrated OpenCV computer vision, LLM parsing pipeline for structured data extraction into large codebase

Columbia Bioelectronic Systems Lab, Research Intern – New York, NY — May 2024 – Aug 2024

- Orchestrated 10 robotic lab devices with 1.0μm precision by writing real-time control software | GitHub Link

Project Experience & Awards

Transformer Accelerator on ESP: Extended HLS4ML compiler with custom C++ translators for matmul and transpose ops, implementing hardware primitives necessary for transformer attention on FPGAs | Project Link

FPGA CNN Hardware Accelerator: Co-designed HW/SW accelerator in SystemVerilog, C integrating quantized TensorFlow CNN (int8, scale/zero-points), achieving 3.4x convolution speedup at 30% utilization | Project Link

Linux Kernel Scheduler & Filesystem: Implemented custom priority scheduler (+75% throughput) and full read/write filesystem in Linux kernel (C), demonstrating OS-level performance/latency trade-offs | GitHub Link

iOS-Based Robot Computer Vision: Real-time object tracking system via TCP network packets | GitHub Link

Apple WWDC Scholarship: 1 of 350 chosen, physics-based game simulating collisions and gravity | GitHub Link

Skills, Languages, & Technologies

Languages: C++, C, Python, Bash, Java, Swift | **ML/AI:** PyTorch, TensorFlow, Transformers, NumPy

Systems: NVIDIA MIG, Linux kernel, HW/SW co-design, profiling, FPGA (Quartus, Vitis HLS), Verilog, SystemC